

STUDENT ATTENDANCE MANAGEMENT

ABSTRACT

Module 1: Student Data Management

This module handles the storage and management of student information such as roll number, name, total working days, and present days. It uses **structures in C** to group related data efficiently and allows easy access and modification of student records.

Module 2: Attendance Calculation & Reporting

This module focuses on calculating the attendance percentage using a function. It processes the input data and generates a formatted attendance report. Arrays and loops are used to handle multiple student records and display results systematically.

INTRODUCTION

The Student Attendance Management System is a mini project developed using the C programming language. It is designed to store student details and calculate their attendance percentage efficiently. The system demonstrates the practical application of basic programming concepts such as structures, functions, arrays, and loops. It helps reduce manual effort in attendance calculation and improves accuracy.

OBJECTIVES

- To store student details in a structured format
 - To calculate attendance percentage automatically
 - To generate and display attendance reports
 - To understand real-world applications of C programming
 - To improve problem-solving and coding skills
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REQUIREMENTS

Software Requirements

- C Compiler (e.g., Turbo C, GCC)
- Code Editor (e.g., VS Code, Code::Blocks)
- Operating System (Windows/Linux)

Hardware Requirements

- Basic Computer System
- Minimum 2GB RAM
- Keyboard and Monitor

ALGORITHM:

1. Start
2. Declare structure `Student` with:
 - roll number
 - name
 - total days
 - present days
3. Input number of students `n`
4. Create an array of structures for `n` students
5. For each student (`i = 1` to `n`):
 - Input roll number
 - Input name
 - Input total working days
 - Input present days
6. For each student:
 - Calculate attendance percentage
 - $\text{percentage} = (\text{present} / \text{total}) \times 100$
7. Display attendance report:
 - Roll No, Name, Present/Total, Percentage
8. Stop

FLOWCHART:

START

|

v

Enter number of students (n)

|

v

i = 0

|

v

Input student details

(roll, name, total, present)

|

v

i = i + 1

|

i < n ?

/ \

YES NO

|

|

v

v

Repeat Calculate Percentage

|
v
Display Report
|
v
STOP

CODE:

```
#include <stdio.h>

struct Student {
    int rollNo;
    char name[50];
    int totalDays;
    int presentDays;
};

/* Function to calculate attendance percentage */
float calculatePercentage(int present, int total) {
    return ((float)present / total) * 100;
}

int main() {
    int n, i;
```

```
printf("Enter number of students: ");
scanf("%d", &n);
struct Student s[n];
// Input student details
for (i = 0; i < n; i++) {
printf("\nEnter details of student %d\n", i + 1);
printf("Roll Number: ");
scanf("%d", &s[i].rollNo);
printf("Name: ");
scanf(" %[^\\n]", s[i].name);
printf("Total Working Days: ");
scanf("%d", &s[i].totalDays);
printf("Total Present Days: ");
scanf("%d", &s[i].presentDays);
}
// Display attendance report
printf("\n----- Attendance Report -----\\n");
printf("Roll No\\tName\\t\\tPresent/Total\\tPercentage\\n");
for (i = 0; i < n; i++) {
float percentage =
calculatePercentage(s[i].presentDays, s[i].totalDays);
```

```
printf("%d\t%s\t\t%d/%d\t\t%.2f%%\n",
s[i].rollNo,
s[i].name,
s[i].presentDays,
s[i].totalDays,
percentage);
}

return 0;
}
```

METHODOLOGY

1. **Data Input:**
User enters the number of students and their details (roll number, name, total days, present days).
2. **Data Storage:**
Use of **structures and arrays** to store multiple student records.
3. **Processing:**
A function calculates attendance percentage using the formula:
$$\text{Percentage} = \frac{\text{Present Days}}{\text{Total Days}} \times 100$$
4. **Output Generation:**
The system displays a formatted attendance report with all student details.
5. **Looping Mechanism:**
Loops are used to repeat input and output operations for multiple students.

REFERENCES

- C Programming Language by **Dennis Ritchie & Brian Kernighan**
- Online tutorials (GeeksforGeeks, TutorialsPoint)
- Class notes and lab manuals
- College lecture materials

